$$
\begin{aligned}
x &= R\sin\theta\cos\phi \\
y &= R\sin\theta\sin\phi \\
z &= R\cos\theta
\end{aligned}
$$

$$
\begin{aligned}
dx &= R(\cos\theta\cos\phi d\theta - \sin\theta\sin\phi d\phi) \\
dy &= R(\cos\theta\sin\phi d\theta + \sin\theta\cos\phi d\phi) \\
dz &= -R\sin\theta d\theta
\end{aligned}
$$

$$
\begin{aligned}
dx^2 &= R^2(\cos^2\theta\cos^2\phi d\theta^2 + \sin^2\theta\sin^2\phi d\phi^2 - 2\cos\theta\sin\theta\cos\phi\sin\phi d\theta d\phi) \\
dy^2 &= R^2(\cos^2\theta\sin^2\phi d\theta^2 + \sin^2\theta\cos^2\phi d\phi^2 + 2\cos\theta\sin\theta\cos\phi\sin\phi d\theta d\phi)
\end{aligned}
$$

$$dx^2 + dy^2 = R^2(\cos^2\theta d\theta^2 + \sin^2\theta d\phi^2)$$

$$
\begin{aligned}
xdx &= R^2(\sin\theta\cos\theta\cos^2\phi d\theta - \sin^2\theta\cos\phi\sin\phi d\phi) \\
ydy &= R^2(\sin\theta\cos\theta\sin^2\phi d\theta + \sin^2\theta\cos\phi\sin\phi d\phi)
\end{aligned}
$$

$$xdx + ydy = R^2\sin\theta\cos\theta d\theta$$

$$dl^2 = R^2(d\theta^2 + \sin^2\theta d\phi^2)$$

$$\theta = \chi$$

$$dl^2 = R^2(d\chi^2 + \sin^2\chi d\phi^2)$$

$$\int dl = 2\pi R\sin\chi$$

$$\int dS = \int_0^{2\pi} d\phi \int_0^{\pi} R^2 \sin\chi d\chi = 4\pi R^2$$

$$\int dS = \int_0^{2\pi} d\phi \int_0^{\chi} R^2 \sin\chi d\chi = 2\pi R^2(1-\cos\chi) = 2\pi R^2(1-\cos\frac{r}{R})$$